

DECISION BRIEF

Quality blockers and funding

What blocks live quality, what to drive next, and whether any new subscription is required. Pay for models only after the index can finish and stay on disk.

System	DriveFaceIndexer + Carousel Studio
Date	17 August 2026
Audience	Product and ops
Source	docs/quality-blockers-and-subscriptions.md

No

New SaaS vendor required

Yes

Fund OpenRouter credits

Keep

Claude + Gemini billed

Pay models after the index can finish. Disk, overlapping video claims, and empty Whisper cues collapse quality before extra LLM spend can show up. Extra quota without isolation shows up as more 429s and cached junk.

1. Quality is five outcomes

Each outcome has a different saturator. A better model cannot fix the wrong one.

Outcome	Quality means	Saturator if it fails
Library permanence	PROCESSED work stays when Drive folders change	Indexer policy + disk
Search	Right faces, scenes, spoken moments rank first	Faces, captions, embeddings, transcripts
Video index	Transcript + faces + frames exist	CPU + disk + claim locks
Carousel script	Hooks people would post	LLM quality + verbatim timestamps
Carousel images	Centered, front-facing guest	Local face scores, then vision

2. Blockers — do these before buying more models

Ranked by what blocks live quality today, not the full architecture register.

ID	Blocker	Why quality dies	Drive this
B1	Leftover Drive downloads refill volume	ENOSPC stops index, preview, backup. Manual cleanup already recovered ~41 GiB.	Auto-unlink on success and ERROR. Keep uploads and YouTube. Do not buy more volume first.
B2	Overlapping video workers + 900s stall watchdog	Proof video committed Media/segments then ERROR. Empty transcript, no VLM cues.	One indexer replica until leases. Fencing token on finalize.

ID	Blocker	Why quality dies	Drive this
B3	Whisper / empty transcript cues	Hooks must be exact transcript sentences. Empty cues make carousels unusable.	Confirm faster-whisper on the indexer image. Backfill empty-cue videos. No new vendor.
B4	API and indexer share one process	Search and InsightFace fight for CPU, RAM, DB pool. Carousel feels broken during video index.	Split dfi-backend vs dfi-indexer (same image). Railway replica, not a SaaS.
B5	Gemini quota mixed with backfill	Caption/embed 429s starve interactive generate. Weak fallbacks get cached.	Pause backfill on 429. Reserve interactive quota before raising the Gemini tier.
B6	Synchronous carousel LLM stages	Timeouts and retries double spend while the provider keeps running.	Persist per-stage jobs + idempotency.
B7	Weak generations poison cache	Cache key is provider+model+transcript, not prompt version. Bad themes stick.	Claude remains script default. Force new cache id on regenerate.
B8	No search-quality regression gate	eval_search_quality.py exists but is not an SLO on deploy.	Run golden set after indexer deploys. Uses existing Gemini key.

Not quality blockers this quarter: Qwen GPU sidecar, Cloud Vision, SerpAPI, Cohesivity, copying /test onto /carousel.

3. Quality ladder

Climb the next rung only when the previous one is green.

Rung	Drive	Done looks like	How
1	Finish the index	Media + non-empty transcript + faces + frames	Local InsightFace, ffmpeg, faster-whisper. Gemini embeddings and captions stay the search backbone.
2	Grounded carousel text	Themes, topics, hooks people would post	Claude Sonnet 4.5 default. Cheap draft then polish once. Hooks must be transcript substrings.
3	Intentional frames	Centered, front-facing guest matching the hook	Local face scores first. Vision only as a tie-breaker.
4	Search relevance loop	Golden queries stay true after indexer deploys	People vs scene vs spoken-text precision, judged with the Gemini key you already have.

4. Accounts

Keep — already the product

Account	Powers	Subscription
Railway	API, carousel, volume, Postgres/Qdrant	Keep. Indexer split may add one replica.
Google Cloud	Drive API, OAuth, Picker	Project + APIs. Not a Face product.
Gemini API	Embeddings, captions, VLM, search eval	Keep paid usage. Quality floor for search.
Anthropic Claude	Default themes / extract / polish	Keep billed. Cheap scripts are the regression.

Fund — only new spend that lifts quality

Account	Unlocks	Need a subscription?
OpenRouter credits	Model picker, Gemini/Claude failover, optional GPT/Llama	Credits, not a required monthly plan. Only new spend that lifts carousel quality and uptime.
Gemini tier bump	Higher embed/caption/VLM concurrency	Only after interactive vs backfill isolation. Raising first fills 429s faster.

OpenRouter does not replace Gemini embeddings or Claude-as-default. Fund credits for generate plus a small failover buffer. Image-gen models can wait.

Do not buy this quarter

Offer	Why skip
Qwen / GPU VLM host	Railway has no GPU. Gemini already captions.
Cloud Vision Web Detection	Repo notes it is weak for people ID.
SerpAPI Google Lens	Skip unless reverse-image identity is a launch gate.
Apify Google Lens	Useful for review-queue naming, not scripts or embeddings.
Cohesivity / Exa	LinkedIn enrichment, not index quality.
Hosted Whisper / Deepgram	faster-whisper is already wired. Pay only if B2/B3 still leave empty cues.
Gemini File Search revival	Retired. Qdrant is the index.
Bigger Railway volume	Treats B1's symptom. Auto-clean first.

YouTube Netscape cookies and the Google Picker HTTP-referrer allowlist are operational config, not paid plans.

5. Work order

- 1 Auto-unlink leftover Drive caches on success/ERROR (B1).
- 2 Single-owner video execution; stop stall-watchdog false ERROR (B2).
- 3 Whisper present + backfill empty transcripts (B3).
- 4 API / indexer service split (B4).
- 5 Quota isolation + OpenRouter credits for generate/failover (B5).
- 6 Golden search eval on deploy (B8).
- 7 Only then: Whisper model size bump, vision tie-breakers, Apify.

Decision. No new SaaS vendor. Keep Claude and Gemini billed. Fund OpenRouter credits. Raise Gemini quota only after backfill is isolated. Skip GPU, Vision, SerpAPI, Apify, and Exa unless people reverse-search becomes a launch gate.

Until rungs 1–3 are green, extra model subscriptions will not show up as better carousels or better search. They will show up as more 429s and more cached junk.